

IMO(17th) – 1975

First Day

- Let $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$ be two n -tuples of numbers. Prove that $\sum_{i=1}^n (x_i - y_i)^2 \leq \sum_{i=1}^n (x_i - z_i)^2$ is true when $z_1, z_2, \dots, z_n$ denote $y_1, y_2, \dots, y_n$ taken in another order.
- Let $a_1, a_2, a_3, \dots$ be any infinite increasing sequence of positive integers. (For every integer $i > 0, a_{i+1} > a_i$.) Prove that there are infinitely many m for which positive integers x, y, h, k can be found such that $0 < h < k < m$ and $a_m = xa_h + ya_k$.
- On the sides of an arbitrary triangle ABC , triangles BPC, CQA and ARB are externally erected such that: $\angle PBC = \angle CAQ = 45^\circ, \angle BPC = \angle QCA = 30^\circ, \angle ABR = \angle BAR = 15^\circ$, Prove that $\angle QRP = 90^\circ$ and $QR = RP$.

Second Day

- Let A be the sum of the digits of the number 4444^{4444} and B is the sum of the digits of the number A . Find the sum of the digits of the number B .
- Is it possible to plot 1975 points on a circle with radius 1 so that the distance between any two of them is a rational number (distances have to be measured by chords)?
- The function $f(x, y)$ is a homogeneous polynomial of the n th degree in x and y . If $f(1, 0) = 1$ and for all a, b, c
 $f(a + b, c) + f(b + c, a) + f(c + a, b) = 0$, then prove that $f(x, y) = (x - 2y)(x + y)^{n-1}$.

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